# Model-Agnostic Feature Selection Technique to Improve the Performance of One-Class Classifiers

*Note: This report includes the implementation results of the study with the same name.

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## I. Introduction

The study presents a model-agnostic feature selection technique to improve the performance of One-Class Classifiers (OCC) useful in areas such as Credit Card Fraud Detection. Traditionally, OCC models are difficult to interpret and understand which features affect the predictions. This study aims to determine the feature importance using SHAP (SHapley Additive exPlanations) technique and perform feature selection based on this importance. In the study, One-Class Gaussian Mixture Models (One-Class GMMs) and One-Class Support Vector Machines (One-Class SVMs) were evaluated on the Credit Card Fraud Detection Dataset.

The results show that One-Class GMM models created with SHAP-based feature selection perform significantly better than One-Class GMM models created without feature selection. For One-Class SVM, it is shown that the model created with the full dataset features must have a robust performance for SHAP feature selection to provide additional gain. The study provides new evidence that SHAP can identify informative features for one-class classifiers.

The source code and dataset are available at our GitHub repository: https://github.com/drvshavva/BLG_527E_project.

## II. Methodology

### A. Dataset

The dataset used in the study is the "Credit Card Fraud Detection Dataset" from Kaggle. The authors refer to this dataset as the "Credit Card" dataset [1].

- **Origin:** Derived from the European credit card transaction data as of September 2013.
- **Content:** Contains 284,807 financial transaction records.
- **Labeling:** Only 492 (0.172%) of the records were labeled as fraudulent. This shows that the dataset is extremely imbalanced, making it suitable for one-class classifiers that focus on modeling normal behavior.
- **Features:** The dataset contains 30 features and a label.
- **Transformation:** The transaction data was transformed via Principal Component Analysis (PCA) to protect the privacy of individuals. Except for the features "time" and "transaction amount", the other features are labeled as V1, V2, ... V28.
- **Unused Feature:** The "time" feature was not used in the study because it can act as a unique identifier and cause models to memorize the training data.
- **Scaling:** Features are scaled to the [0,1] range before being input to the classifiers for training or inference

### B. Algorithmns

The study used two popular One-Class Classifiers along with SHAP technique:

**SHAP (SHapley Additive exPlanations):**

- A technique introduced by Lundberg and Lee in a 2017 paper.
- It is used to interpret model predictions.
- It utilizes Shapley values from Game Theory to determine the contribution of a feature to the model output.
- It is model-agnostic, meaning it can be applied to any type of model.
- It has three desirable properties, namely local accuracy, incompleteness, and consistency.
- Feature importance values (SHAP values) can be positive or negative real numbers; their absolute values indicate the importance of the feature. The absolute values of the SHAP values of the samples can be averaged to obtain an aggregate feature importance for a dataset.
- It is available as an open-source library for the Python programming language (shap library [2]).
- In this study, Lundberg and Lee's first formulation, Kernel SHAP, was used.

**One-Class Support Vector Machine (One-Class SVM):**

- The Sklearn implementation works as described by Schölkopf et al.
- It can be viewed as a specialization of SVM that can be trained on data where all examples have the same label.
- Unlike the binary classification SVM, which finds a maximum marginal separation hyperplane that is maximally far from the boundaries of the regions containing examples of the two classes, the One-Class SVM finds the maximum marginal separation hyperplane between the training data and the origin.
- The Radial Basis Function (RBF) kernel is used in the study. With the RBF kernel, the One-Class SVM can be

conceptualized as finding a surrounding hypersphere that contains most of the examples of the training data. The test examples are classified according to whether they are inside this hypersphere or not.

- In the study, the nu parameter was set to 0.0025, the kernel parameter "rbf" and the gamma parameter to 0.0310 (hyperparameters were selected by grid search).
- Since the Sklearn implementation does not provide direct probability estimation, sigmoid calibration was applied to obtain the class probabilities required for the performance metric (AUPRC).

**One-Class Gaussian Mixture Model (One-Class GMM):**

- The Sklearn implementation uses the expectation-maximization algorithm to fit a set of Gaussian distributions to the data for classification.
- It models the training data, which contains examples belonging to a single class, as a mixture or sum of multivariate Gaussian distributions.
- During training, GMM learns the mean and covariance parameters that best describe the data.
- To perform classification, it calculates the probability that a new test example is generated by the distributions fitted to the training data. If the calculated probability falls below a certain level, the example is identified as belonging to the class not used for training (i.e., anomaly).
- In the study, the n_components parameter is set to 2, and the other hyperparameters are left at their default values (hyperparameters are selected by grid search).
- It was found that the probabilities of the predict_proba method of the sklearn implementation were concentrated near 0.0 or 1.0 and did not provide enough points to generate a meaningful precision-recall curve. Therefore, sigmoid calibration was applied to obtain a more finely spaced and interpretable precision-recall curve.

### C. *Experiment*

The method of the study was implemented as a two-phase process (Figure 1). Each phase was carried out by running a Python program.

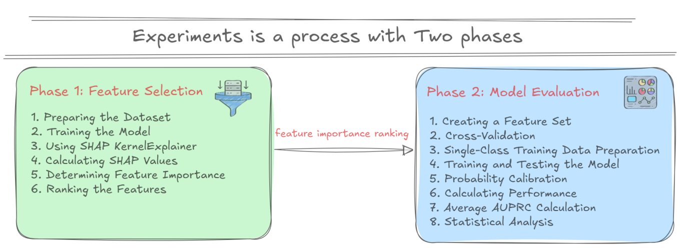


Fig. 1: Two Phase Experiment

**Phase 1: Feature Selection:** The purpose of this phase is to rank the features of the dataset according to their importance using SHAP values.

1) **Preparing the Dataset:** The Credit Card dataset is used. 29 features are used except for the "time" feature. The features are scaled to the range [0,1].
2) **Training the Model:** For feature selection, an uncalibrated One-Class SVM or One-Class GMM model is trained on the entire dataset. The model's hyperparameters are fixed (n_components=2 for GMM, nu=0.0025 for SVM, kernel="rbf", gamma=0.0310).
3) **Using SHAP KernelExplainer:** The trained model and a sample of the dataset are given to the SHAP KernelExplainer object. A default of 100 samples are taken using the shap.sample function.
4) **Calculating SHAP Values:** The shap_values function of the KernelExplainer object is called. This function returns a two-dimensional array of SHAP (feature importance) values for each sample.
5) **Determining Feature Importance:** The feature importance value of each feature is found by averaging the absolute values of each row of the SHAP array (i.e. each sample). This technique is also used by Lundberg in the documentation of the shap library.
6) **Ranking the Features:** The features are ranked according to their calculated average absolute SHAP values. This ranking is done from the feature with the highest SHAP value to the lowest. The results of this step are lists that give the order of importance of the features.

**Phase 2: Model Evaluation:** In this phase, models are created using different numbers of features according to the feature importance ranking determined in Phase 1 and their performances are evaluated.

1) **Feature Set Creation:** According to the feature importance ranking created in Phase 1, subsets consisting of the first n features with the highest SHAP importance are determined. In the study, values of n are used as 3, 5, 7, 10 and 15. In addition, a case where all 29 features are used is included for comparison.
2) **Data Splitting (Cross-Validation):** Ten repetitions (ten rounds) of five-fold cross-validation are used for model evaluation. The data is divided into training and test subsets by setting k=5 with scikit-learn's k-fold cross validation library.
3) **Single Class Training Data Preparation:** Since we are working with single class classifiers, after the data is divided into training and test subsets, minority class (anomaly) examples in the training data are removed. The class distribution in the test subset is not changed.
4) **Training and Testing the Model:** One-Class SVM and One-Class GMM models are trained for each feature subset (3, 5, 7, 10, 15 and 29 features). The hyperparameters of the model are the same as in Phase 1.
5) **Probability Calibration:** The output (decision scores) of the One-Class SVM and One-Class GMM models are subjected to sigmoid calibration to obtain the class probabilities required for the AUPRC metric.
6) **Calculating Performance:** The performance of the

model for each test fold is calculated and recorded using the Area Under the Precision Recall Curve (AUPRC) metric. AUPRC measures the balance between precision and recall and is especially suitable for imbalanced data sets. A higher AUPRC value means better performance.

7) **Average AUPRC Calculation:** Since ten repeats of five-fold cross validation are performed, a total of 50 AUPRC scores are recorded for each model. 50 scores are obtained for each number of features (6 different values), resulting in a total of 300 AUPRC scores for each classifier. The average of these 50 scores is taken to obtain the average AUPRC performance for each number of features.
8) **Statistical Analysis:** Statistical analysis is performed on the average AUPRC results. ANOVA (Analysis of Variance) test is used to determine the effect of the number of features on the experimental results. If the p-value as a result of the ANOVA test is below the predetermined significance level (0.01), it is concluded that the number of features has a significant effect. In this case, Tukey's Honestly Significant Difference (HSD) test is applied to rank the effects of different number of features on performance. The HSD test ranks the number of features into groups according to their effects on AUPRC scores.

## III. Results

### A. Performance Metric: AUPRC

The Area Under the Precision Recall Curve (AUPRC) is used for model performance in the study.

**Why Used?** Previous studies have shown that AUPRC is a better metric than other common metrics (e.g. AUC - Area Under the Receiver Operating Characteristic Curve), especially for highly imbalanced data classification. The Credit Card dataset is extremely imbalanced.

**How is it calculated?** Precision and Recall are plotted for different classification thresholds and the area under the curve is calculated.

The AUPRC value is between zero and one; a higher AUPRC value indicates better performance.

### B. Reproduction of Experiments - SHAP Feature Importance

In this study, feature selection was performed using the SHAP (SHapley Additive exPlanations) method. The process began by training One-Class GMM and One-Class SVM models on the full dataset. SHAP was then applied to the trained models to calculate feature importance scores, and subsets of features were selected based on their ranked importance. The models were retrained using these selected feature subsets, and their outputs were evaluated. The results obtained for GMM and SVM are presented below.

During the re-implementation, SHAP-based feature importance rankings were compared with those in the original study. A total of 8 common features were identified for the SVM model, and 12 common features were found for the GMM model. These shared features are highlighted in bold in the comparison tables.

TABLE I: Comparison of features selected with SHAP for SVM model

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Paper | **V1** | V23 | **V25** | **V3** | **V18** | V28 | V2 | **V19** |
| | V21 | **V15** | V5 | V20 | **V4** | **V11** | V16 | |
| Implementation | V13 | **V4** | V26 | V24 | **V1** | **V15** | **V11** | **V19** |
| | **V18** | **V3** | V12 | V9 | V14 | V22 | **V25** | |

TABLE II: Comparison of features selected with SHAP for GMM model

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Paper | V11 | **V2** | V26 | **V12** | **V8** | **V5** | **V9** | V7 |
| | **V18** | **V3** | **V10** | **V16** | **V17** | **V21** | **V14** | |
| Implementation | **V8** | **V3** | **V2** | **V17** | **V21** | **V5** | **V14** | **V12** |
| | **V16** | **V18** | 22 | **V10** | V13 | V6 | **V9** | |

To evaluate the impact of the number of selected features on anomaly detection performance, One-Class GMM and One-Class SVM models were trained using feature subsets of varying sizes. For each feature count, the average AUPRC (Area Under the Precision-Recall Curve) score was calculated across 50 repeated runs to account for statistical variability. The results obtained from the re-implementation were compared against those reported in the original study.

TABLE III: Comparison of mean AUPRC scores for One-Class GMM model

| #Features | Paper | Impl. (mean ± std) |
|---|---|---|
| 3 | 0.2515 | 0.1232 ± 0.0311 |
| 5 | 0.3669 | 0.3989 ± 0.0391 |
| 7 | 0.3411 | 0.4563 ± 0.0408 |
| 10 | 0.4071 | 0.4841 ± 0.0381 |
| 15 | 0.6420 | 0.5585 ± 0.0382 |
| 29 | 0.4971 | 0.4822 ± 0.0409 |

TABLE IV: Comparison of mean AUPRC scores for One-Class SVM model

| #Features | Paper | Impl. (mean ± std) |
|---|---|---|
| 3 | 0.0114 | 0.0524 ± 0.0159 |
| 5 | 0.1748 | 0.0268 ± 0.0097 |
| 7 | 0.1644 | 0.1310 ± 0.0304 |
| 10 | 0.1429 | 0.2272 ± 0.0391 |
| 15 | 0.2835 | 0.3703 ± 0.0428 |
| 29 | 0.3775 | 0.4024 ± 0.0426 |

As shown in the tables, the AUPRC scores obtained in this implementation differ numerically from those in the original paper, but follow similar performance trends. For both GMM and SVM models, feature subset sizes of 15 and 29 consistently yielded higher average AUPRC scores, indicating better detection capability with more informative features. While GMM performance improved steadily with increasing feature count up to 15, the SVM model showed a sharper performance jump beyond 10 features. These results reinforce the importance of feature selection in one-class classification tasks, while also highlighting some sensitivity to implementation-level variations.

ANOVA (Analysis of Variance) is used to test whether different levels of an independent variable have a statistically significant effect on model performance. In this study, ANOVA was applied to examine the impact of the number of features on the AUPRC performance of the One-Class GMM and SVM models.

TABLE V: ANOVA results for One-Class GMM model

| Source | Df | Sum Sq | Mean Sq | F value | Pr(¿F) |
|---|---|---|---|---|---|
| Paper | | | | | |
| Features | 5 | 4.64 | 0.93 | 264.16 | * |
| Residuals | 294 | 1.03 | 0.00 | | |
| Impl. | | | | | |
| Features | 5 | 5.85 | 1.17 | 802.10 | * |
| Residuals | 294 | 0.43 | 0.00 | | |

TABLE VI: ANOVA results for One-Class SVM model

| Source | Df | Sum Sq | Mean Sq | F value | Pr(¿F) |
|---|---|---|---|---|---|
| Paper | | | | | |
| Features | 5 | 3.94 | 0.79 | 718.36 | * |
| Residuals | 294 | 0.32 | 0.00 | | |
| Impl. | | | | | |
| Features | 5 | 6.36 | 1.27 | 1183.38 | * |
| Residuals | 294 | 0.32 | 0.00 | | |

Tukey's HSD (Honestly Significant Difference) test is applied after an ANOVA to identify which specific groups differ significantly from each other. It performs pairwise comparisons to find which levels of a factor lead to statistically different outcomes. This helps determine at which feature counts the model performance varies meaningfully.

TABLE VII: Comparison of HSD test groupings for One-Class GMM AUPRC scores

| **Study** | **HSD Test Groupings** |
|---|---|
| Paper | Group a consists of: 15<br>Group b consists of: 29<br>Group c consists of: 10<br>Group cd consists of: 5<br>Group d consists of: 7<br>Group e consists of: 3 |
| Implementation | Group a consists of: 15<br>Group b consists of: 7, 10, 29<br>Group c consists of: 5<br>Group d consists of: 3 |

TABLE VIII: Comparison of HSD test groupings for One-Class SVM AUPRC scores

| **Study** | **HSD Test Groupings** |
|---|---|
| Paper | Group a consists of: 29<br>Group b consists of: 15<br>Group c consists of: 5<br>Group cd consists of: 7<br>Group d consists of: 10<br>Group e consists of: 3 |
| Implementation | Group a consists of: 15, 29<br>Group b consists of: 10<br>Group c consists of: 7<br>Group d consists of: 3, 5 |

## DISCUSSION

During the re-implementation process, the methodology presented in the original paper was closely followed. However, due to differences in computational environment, software versions, random seed initialization, and possible variations in data preprocessing or sampling, the numerical results do not exactly match those reported in the original study.

For instance, in the ANOVA results, the F value for the One-Class GMM model in this implementation was considerably higher (802.10) compared to the original paper (264.16). A similar pattern was observed for the SVM model, where the F value also increased significantly (1183.38 vs. 718.36). These differences may stem from variations in SHAP value estimation, such as differences in approximation precision, feature correlation effects, or random subsampling procedures.

Additionally, discrepancies were found in the HSD (Honestly Significant Difference) test groupings. In this implementation, some feature counts were grouped differently or appeared in multiple overlapping groups, suggesting a greater dispersion in model performance across different feature subset sizes. Nevertheless, the overall trends remained consistent—for example, feature set size 15 consistently appeared in the top-performing group in both GMM and SVM models.

These deviations, while notable in their magnitude, do not contradict the original paper's core findings; instead, they highlight the sensitivity of model performance to subtle implementation-level factors in model interpretation and selection pipelines.

## REFERENCES

[1] M. Dal Pozzolo, "Credit Card Fraud Detection Dataset," *Kaggle*, 2015. [Online]. Available: https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

[2] S. Lundberg *et al.*, "SHAP," *GitHub*, Available: https://github.com/slundberg/shap/tree/v0.41.0, accessed: Jul. 9, 2023.